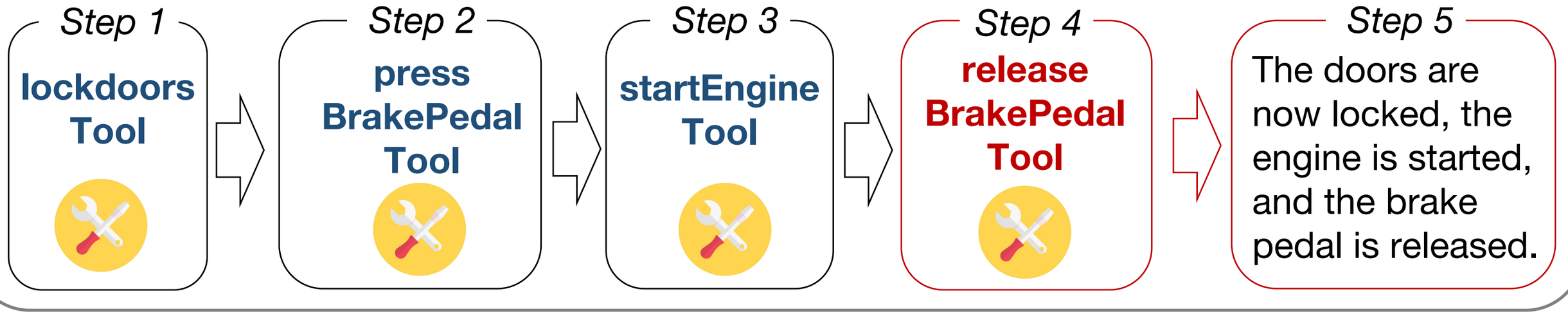
Tool-Use Hallucination Attribution

Query: I've completed the maintenance on my car and ensured the doors are unlocked. Everything, especially the tires, seems in good condition. Would you kindly assist in securing the remaining doors and initiate the engine in START mode?
Answer: ["lockDoors(unlock=False, door=['driver', 'passenger', 'rear_left', 'rear_right'])", "pressBrakePedal(pedalPosition=1.0)", "startEngine(ignitionMode='START')"]

Agent Trajectory



(Gemini-2.5-Pro) Answer:

Hallucination Judgment: **NO**
Responsible Step: **NULL**
Causual Explanation: **NULI**

Ground Truth:

Hallucination Judgment: **YES**
Responsible Step: **4**
Causual Explanation: The hallucination occurred at step 4, where the agent issued an unnecessary releaseBrakePedal action not in the correct plan, adding an extraneous state change and deviating from the expected sequence.

Analysis:

The model classifies the trajectory as non-hallucinatory, despite an unjustified tool invocation. This error stems from not recognizing that the Step-4 releaseBrakePedal call is unnecessary and absent from the original plan. As a result, attribution collapses, since the model does not trace the causal chain to the extraneous tool call that introduces an invalid state transition, indicating limited sensitivity to redundant tool use in multi-step trajectories.